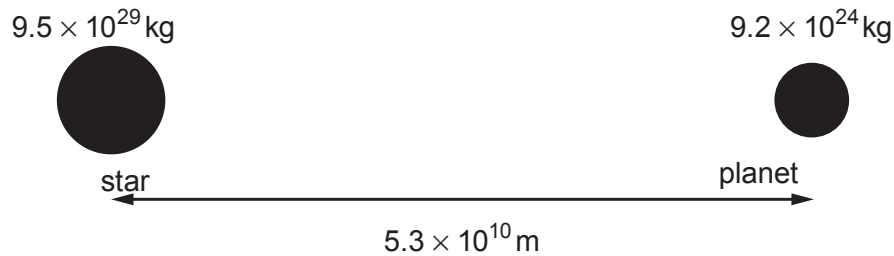


3. The planet (Kepler-186f) orbits the star (Kepler-186) and is considered to be the most Earth-like planet yet discovered. Approximate details of the system are shown in the diagram below.



- (a) (i) Show that the radius of orbit of the star around the centre of mass of the system is 513 km. [2]

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- (ii) The period of orbit of the planet is 130 days. Calculate the orbital velocity of the star. [3]

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- (iii) The velocity calculated in part (a)(ii) is small. Explain why small velocities are difficult to measure using red shift. [2]

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- (b) (i) Calculate the gravitational force between the star and the planet. [2]

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- (ii) An historic alternative to Newton's Law of Gravitation involved invisible steel rods connecting stars and planets and providing the force for keeping planets in orbit. A hypothetical cylindrical steel rod of radius $7.0 \times 10^6 \text{ m}$ and length $5.3 \times 10^{10} \text{ m}$ joins the planet to the star. Determine whether or not this steel rod is strong enough to keep the planet in orbit and explain why this theory is not accepted. (Breaking stress of steel = $4.5 \times 10^8 \text{ Pa}$.) [4]

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- (c) (i) Calculate the potential energy of the star-planet system (see diagram in part (a)). [2]

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- (ii) In fact the planet has a slightly elliptical orbit. Explain how conservation of energy applies to this elliptical orbit. [2]

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